

Enhancing Federated Class-Incremental Learning via Spatial-Temporal Statistics Aggregation

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Abstract

The growing presence of mobile and IoT devices has led to massive decentralized and evolving data, driving the rise of Federated Learning (FL) to enable collaborative training without data sharing. However, traditional FL assumes static data distributions, which is unrealistic for dynamic real-world environments. To address this challenge, Federated Class-Incremental Learning (FCIL) has emerged as a promising framework that enables flexible adaptation to newly introduced classes over time. Existing FCIL methods typically integrate old knowledge preservation into local client training. However, these methods cannot avoid spatial-temporal client drift caused by data heterogeneity and often incur significant computational and communication overhead, limiting practical deployment. To address these challenges simultaneously, we propose a novel approach, Spatial-Temporal Statistics Aggregation (STSA), which provides a unified framework to aggregate feature statistics both spatially (across clients) and temporally (across stages). The aggregated feature statistics are unaffected by data heterogeneity and

can be used to update the classifier in closed form at each stage. Additionally, we introduce STSA-E, a communication-efficient variant that enables the server to approximate global second-order feature statistics using first-order statistics uploaded from clients. Theoretical analysis shows that it achieves similar performance to STSA with much lower communication overhead. Extensive experiments on three widely used FCIL datasets, with varying degrees of data heterogeneity, show that our method outperforms state-of-the-art FCIL methods in terms of performance, flexibility, and both communication and computation efficiency. The code is available at <https://github.com/Yuqin-G/STSA>.

CCS Concepts

• **Computing methodologies** → **Distributed computing methodologies**.

Keywords

Federated Learning, Federated Continual Learning, Feature Statistics, Data Heterogeneity

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1 Introduction

In recent years, the rapid proliferation of mobile and IoT devices has generated massive volumes of decentralized data across heterogeneous environments. This trend has spurred the development of Federated Learning (FL) systems [19, 30], which enable collaborative model training without sharing raw data, thereby ensuring privacy and reducing data transmission overhead. However, most existing FL systems assume static data distributions, which is unrealistic for dynamic Web, mobile, and Web of Things (WoT) environments, where user behaviors and data continually evolve over time [10, 40]. To address FL tasks in dynamic environments, recent studies have integrated Continual Learning (CL) techniques into the FL framework, forming the paradigm known as Federated Continual Learning (FCL) [10, 25, 40]. Among its variants, Federated Class-Incremental Learning (FCIL) [4, 15, 37, 41] has attracted increasing attention for its broad applicability [20, 41], as it enables the flexible incorporation of new classes at any time. In this work, we focus on the widely adopted global FCIL [9, 10, 15, 37, 41] setting, where a single global model is collaboratively trained across clients to continually learn new classes, thereby enhancing its generalization across all encountered classes.

In FCIL, we need to simultaneously address two core challenges: catastrophic forgetting and data heterogeneity. Existing FCIL methods typically follow the paradigm where the retention of old knowledge is integrated directly into local client training through techniques such as experience replay [3, 9, 10, 24, 34, 37, 41], weight regularization [26, 36], and parameter-efficient fine-tuning [15, 28] (as shown in Figure 1). However, this paradigm still suffers from two critical drawbacks that limit its practical deployment. ① **Spatial-temporal client drift** caused by data heterogeneity. Specifically, each client has a distinct data distribution, leading to different balances in the optimization process between learning new tasks and preserving old knowledge. This disparity causes inconsistent local updates, degrading the aggregated global model's generalizability to both new and old tasks. As shown in Figure 2, we present the loss landscapes of the local and global models. The local model performs well on old tasks and current local tasks, but the global model shows a noticeable drift with the optimal model. ② **High communication and computation overhead**. Existing methods often necessitate iterative local updates and numerous communication rounds between clients and the server. This imposes significant computational burdens on client devices and leads to high communication overhead, posing practical limitations, especially for resource-constrained clients. These challenges motivated us to explore: *whether it is possible to break this paradigm and design a new method that effectively mitigates client drift caused by data heterogeneity while maintaining low computational and communication overhead*.

In this paper, we propose a method named **Spatial-Temporal Statistics Aggregation (STSA)**. Instead of using aggregated local updates to update the global model, STSA aggregates feature statistics spatially (across clients) to learn new tasks and temporally (across stages) to retain old knowledge, using a fixed global feature extractor. Specifically, in the first stage of STSA, all clients engage in federated training to obtain a global model. The trained backbone

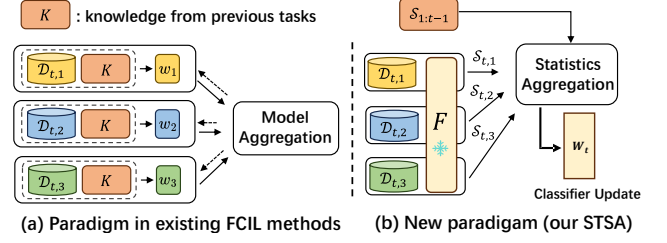


Figure 1: Paradigms of existing FCIL methods and ours. w and S denote the model weights and feature statistics, respectively.

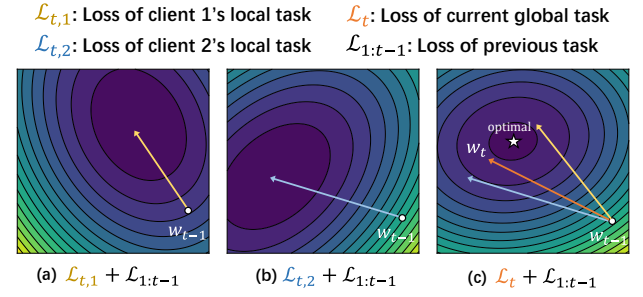


Figure 2: Illustration of spatial-temporal client drift through loss landscapes. w_t denotes the aggregated global model.

is then fixed and distributed among all clients for feature statistics extraction at each subsequent stage. These statistics are first aggregated across clients and then across stages. Notably, the aggregated feature statistics are equivalent to those derived from all data samples across both dimensions due to their linear characteristics, so **they are not affected by data heterogeneity**. Finally, these aggregated statistics are used to update the classifier in closed form at each stage. Since our method **requires no training** after the first stage and only needs one round of communication to upload feature statistics, it significantly improves computation and communication efficiency. Moreover, we propose a more communication-efficient variant, termed STSA-E, which enables the server to approximate global second-order feature statistics based on first-order statistics transmitted from clients. Theoretical analysis shows that STSA-E achieves similar statistical performance to STSA while considerably reducing communication overhead.

Extensive experiments on three widely used FCIL datasets demonstrate the superiority of our methods. In terms of performance, our STSA achieves over **10%** absolute improvements compared to the best-performing parameter-efficient fine-tuning baseline. Moreover, STSA achieves remarkable efficiency improvements, drastically reducing both communication and computation overhead. In particular, it delivers nearly **100×** faster computation compared to existing methods.

In summary, the main contributions of this paper are three-fold:

- We propose a novel FCIL method called STSA, which offers a unified way to aggregate feature statistics both spatially (across clients) and temporally (across stages). The aggregated feature statistics are immune to data heterogeneity and can be used to update the classifier in closed form at each stage.

- We propose STSA-E, a communication-efficient variant where the server approximates global second-order feature statistics using first-order statistics sent by clients. Theoretical and experimental results show that it achieves comparable performance to STSA while significantly reducing communication overhead.
- We conduct extensive experiments to validate our method. The results demonstrate significant improvements in performance, flexibility, and both communication and computational efficiency compared to state-of-the-art FCIL methods.

2 Related Work

Class-Incremental Learning. Continual learning (CL) enables a learning system to acquire knowledge from sequentially arriving tasks while mitigating catastrophic forgetting [46]. Among the various CL settings, Class-Incremental Learning (CIL) is the most widely studied due to its strong relevance to real-world applications [46]. In CIL, each new task introduces additional classes to the output space, leading to a progressively growing set of total classes. Existing CIL methods tackle catastrophic forgetting through strategies such as data replay [2], weight regularization [1, 32], and knowledge distillation [27]. Although current CIL methods have made significant progress, they often struggle to perform well in FCIL. This is due to the new challenges introduced by federated setting, such as data heterogeneity, communication constraints, and privacy concerns, which can exacerbate forgetting and hinder effective knowledge consolidation.

Federated Class-Incremental Learning. GLFC [10] is the first to introduce the FCIL setting and utilizes stored historical exemplars for local relation distillation. Various strategies are then proposed to make the best use of historical exemplars to mitigate catastrophic forgetting [9, 24]. However, storing data from previous tasks is prohibited due to strict privacy regulations and policies in many privacy-sensitive scenarios, e.g., hospitals and medical research institutions [37, 41]. Therefore, the focus has shifted to more challenging exemplar-free scenarios. Existing state-of-the-art exemplar-free FCIL methods can be broadly classified into two categories: (1) Data-Free Knowledge Distillation (DFKD) approaches [3, 31, 34, 37, 41] and (2) Parameter-Efficient Fine-Tuning (PEFT) approaches [4, 15, 28, 44]. DFKD-based FCIL methods train a generator to generate synthetic samples based on the predictions of the global model. These synthetic samples are stored for training subsequent tasks. However, storing and transferring large volumes of synthetic data between the server and clients would incur significant storage and communication costs. PEFT-based FCIL methods store knowledge from previous stages in prompt [4, 28] or LoRA [15] parameters and retain it using similarity matching or task arithmetic. However, these methods fail to fully exploit the generalization capabilities of pre-trained models [45] and remain vulnerable to spatial-temporal client drift, ultimately resulting in poor performance under high data heterogeneity. Besides, they suffer from computational inefficiency due to iterative backpropagation steps, placing a significant burden on client-side resources.

3 Method

3.1 Problem Definition

In FCIL, we have a set of T tasks $\mathcal{T} = \{\mathcal{T}_t\}_{t=1}^T$ shared among K clients, where the set of classes C_t ($|C_t| = c_t$) available at each time

step t is disjoint. Each client k only have access to private dataset $\mathcal{D}_{t,k}$ with size $n_{t,k}$ at task $\mathcal{T}_t$. Here we denote $\mathbf{n}_{t,k} \in \mathbb{R}^{c_t}$ as the label frequency. During the training of task $\mathcal{T}_t$, each local client k trains its local model $\theta_{t,k}$ using its local data $\mathcal{D}_{t,k}$ and then uploads the updated model parameters to the server. The server aggregates the local models to obtain the global model θ_t and distribute it to all clients as the initialization for the next round of training. This process is repeated for multiple rounds, and the final global model θ_t can distinguish the samples belonged to the classes set $\bigcup_{i=1}^t C_i$.

3.2 Spatial-Temporal Statistics Aggregation

First Stage Adaptation. At task $\mathcal{T}_1$, each client participates in federated training to obtain a global model θ_1 . Any federated learning algorithm can be used in this phase. After training, we freeze the feature extractor F from the global model θ_1 for further use.

Spatial Statistics Aggregation. For each task $\mathcal{T}_t$ ($t > 0$), each client k uses this fixed feature extractor F to extract features from its local dataset. Given that the extracted features may lack sufficient expressiveness for clear class separation, we apply a nonlinear random mapping to project them into a higher-dimensional space based on Cover's theorem [7] (from d to M , $M > d$):

$$\mathbf{X}_{t,k} = [\Phi(x_1), \Phi(x_2), \dots, \Phi(x_{n_{t,k}})]^T, \quad (1)$$

where $\{(x_i, y_i)\}_{i=1}^{n_{t,k}} = \mathcal{D}_{t,k}$, $\Phi = R \circ F$ represents the composition of the feature extractor F and the random mapping layer R , and $\mathbf{X}_{t,k} \in \mathbb{R}^{n_{t,k} \times M}$ denotes the concatenation of all feature vectors. Here, R consists of a random matrix followed by a ReLU function. This R is fixed after initialization and shared across all clients. To reduce storage overhead in the clients, the server can distribute a common seed to all clients. This allows clients to generate the same random matrix locally using the seed. After obtaining high-dimensional features, clients can delete it and regenerate it when needed.

Subsequently, each client computes the local gram matrix $\mathbf{G}_{t,k} = \mathbf{X}_{t,k}^T \mathbf{X}_{t,k} \in \mathbb{R}^{M \times M}$ and the feature-label correlation matrix $\mathbf{C}_{t,k} = \mathbf{X}_{t,k}^T \mathbf{Y}_{t,k} \in \mathbb{R}^{M \times c_t}$. Here $\mathbf{Y}_{t,k}$ denotes the corresponding label matrix stacked by one-hot vectors. The Spatial Statistics $\mathbf{S}_{t,k} = \{\mathbf{G}_{t,k}, \mathbf{C}_{t,k}\}$ of each client k is then uploaded to the server and aggregated as follows: $\mathbf{G}_t = \sum_{k=1}^K \mathbf{G}_{t,k}$, $\mathbf{C}_t = \sum_{k=1}^K \mathbf{C}_{t,k}$. Due to the linear additivity of these statistics, we have:

$$\begin{aligned} \mathbf{G}_{\text{centralized}} &= \sum_{(x,y) \in \mathcal{D}_t} \Phi(x)^T \Phi(x) \\ &= \sum_{k=1}^K \sum_{(x,y) \in \mathcal{D}_{t,k}} \Phi(x)^T \Phi(x) = \sum_{k=1}^K \mathbf{G}_{t,k} = \mathbf{G}_t. \\ \mathbf{C}_{\text{centralized}} &= \sum_{(x,y) \in \mathcal{D}_t} \Phi(x)^T \text{OneHot}(y) \\ &= \sum_{k=1}^K \sum_{(x,y) \in \mathcal{D}_{t,k}} \Phi(x)^T \text{OneHot}(y) = \sum_{k=1}^K \mathbf{C}_{t,k} = \mathbf{C}_t \end{aligned} \quad (2)$$

This demonstrates that the statistics derived from Spatial Statistics Aggregation are consistently equivalent to those from the centralized setting (with access to all data from all clients at stage t). Therefore, these statistics are invariant to varying levels of data heterogeneity.

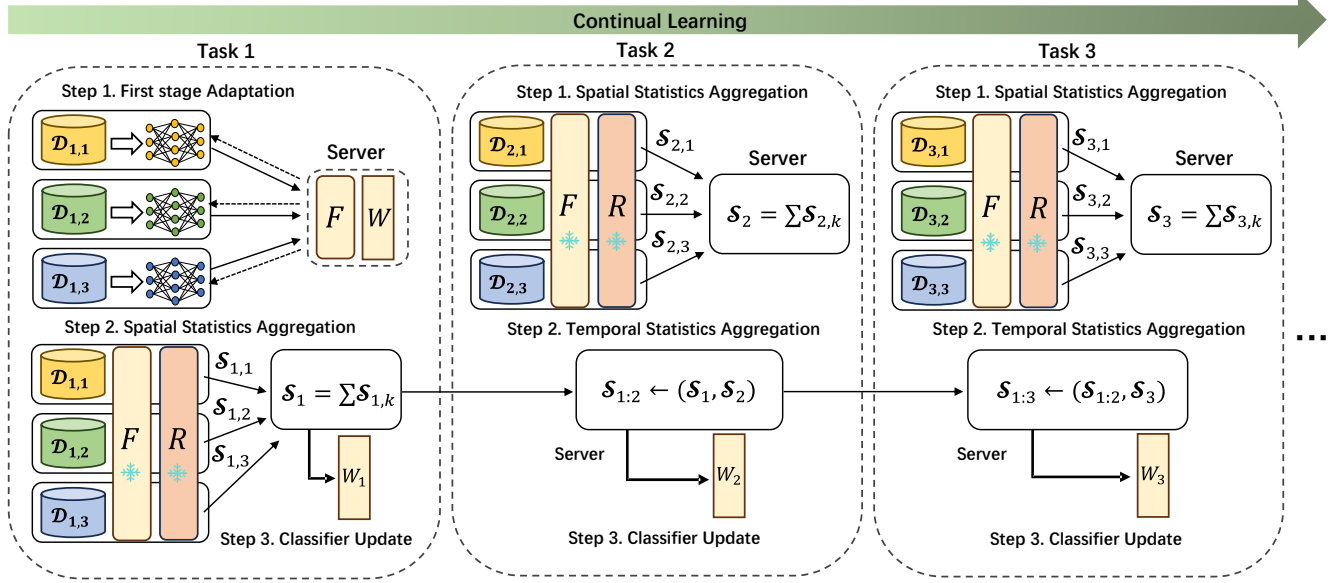


Figure 3: Framework of our algorithm. $\mathcal{D}_{i,j}$ denotes the local dataset of client j for task $\mathcal{T}_i$. F is the feature extractor trained in the first stage and is kept fixed afterward. R is the random mapping layer. In our STSA, Aggregated Spatial Statistics $\mathcal{S}_t$ at stage t includes both G_t and C_t , whereas for STSA-E, it refers to C_t .

Temporal Statistics Aggregation. The server then aggregates the Aggregated Spatial Statistics $\mathcal{S}_t = \{G_t, C_t\}$ with Temporal Statistics $\mathcal{S}_{1:t-1} = \{G_{1:t-1}, C_{1:t-1}\}$, respectively:

$$G_{1:t} = G_{1:t-1} + G_t, \quad C_{1:t} = [C_{1:t-1}, C_t], \quad (3)$$

where $G_{1:t-1} = \sum_{i=1}^{t-1} G_i$ and $C_{1:t-1} = [C_1, C_2, \dots, C_{t-1}]$ are accumulated along $1 \sim (t-1)$ incremental stages. $G_{1:t}$ and $C_{1:t}$ are the latest Temporal Statistics need to store for further use. Given the linear additivity of these statistics, we have:

$$\begin{aligned} G_{\text{joint}} &= \sum_{(x,y) \in \mathcal{D}_{1:t}} \Phi(x)^\top \Phi(x) = \sum_{i=1}^t \sum_{(x,y) \in \mathcal{D}_i} \Phi(x)^\top \Phi(x) = G_{1:t} \\ C_{\text{joint}} &= \sum_{(x,y) \in \mathcal{D}_{1:t}} \Phi(x)^\top \text{One-Hot}(y) \\ &= \sum_{i=1}^t \sum_{(x,y) \in \mathcal{D}_i} \Phi(x)^\top \text{One-Hot}(y) = C_{1:t}. \end{aligned} \quad (4)$$

This indicates that the statistics derived through Temporal Statistics Aggregation are consistently equivalent to those obtained in the joint setting (with access to all data from task $\mathcal{T}_1$ to task $\mathcal{T}_t$).

Classifier Update. After the first stage adaptation, the backbone remains fixed, and only the classifier is updated. The objective of our STSA is to build a strong classifier capable of accurately distinguishing all previously encountered classes. We formulate the solving procedure of the classifier as a regularized least squares form (ridge regression):

$$\min_W \|X_{\text{joint}}W - Y_{\text{joint}}\|_F^2 + \gamma \|W\|_F^2. \quad (5)$$

Here, X_{joint} represents the complete feature matrix, Y_{joint} represents the label matrix of all data samples from tasks $\mathcal{T}_1$ to $\mathcal{T}_t$, γ

controls the strength of the regularization. The closed-form solution of Eq. 5 is:

$$W^* = (G_{\text{joint}} + \gamma I)^{-1} C_{\text{joint}} \quad (6)$$

Based on Eq. 4, we have:

$$\begin{aligned} W_t &= (G_{1:t} + \gamma I)^{-1} C_{1:t} \\ &= (G_{\text{joint}} + \gamma I)^{-1} C_{\text{joint}} = W^*. \end{aligned} \quad (7)$$

Through our Spatial-Temporal Statistics Aggregation, we can obtain the feature statistics required for ridge regression on all data at any stage. Finally, the global model at the end of each stage t is $\theta_t = \{F, R, W_t\}$.

Discussion. Built on the widely observed finding that the feature extractor trained in the first stage has a certain level of generalizability to new classes [29, 45, 46], our STSA extends online least squares [6, 12, 13, 29, 48] to the FCIL scenario by providing a unified way to aggregate feature statistics across both temporal and spatial dimensions. In any scenario with restricted access to the complete dataset, the required statistics for ridge regression on the full dataset can be effectively obtained through our aggregation scheme.

3.3 A communication-efficient variant (STSA-E)

The communication overhead of uploading $G_{t,k}$ ($M \times M$) increases significantly as M grows. To address this, we propose a communication-efficient variant of STSA, named STSA-E. It enables the server to approximate global second-order feature statistics G_t using first-order statistics uploaded from clients.

Proposition 3.1 For each task $\mathcal{T}_t$,

$$\hat{G}_t = \sum_{i=1}^C \left[\frac{n_t^{(i)} - 1}{K - 1} \sum_{k=1}^K \frac{(C_{t,k}^{(i)} \mathbf{1}_c)(C_{t,k}^{(i)} \mathbf{1}_c)^\top}{n_k^{(i)}} - \frac{n_t^{(i)} - K}{n_t^{(i)}(K - 1)} \left(\sum_{k=1}^K C_{t,k}^{(i)} \mathbf{1}_c \right) \left(\sum_{k=1}^K C_{t,k}^{(i)} \mathbf{1}_c \right)^\top \right]. \quad (8)$$

is an unbiased plug-in estimator of the global gram matrix G_t when the global feature set $\Omega = \{(\Phi(x), y) \mid (x, y) \in \cup_{k=1}^K \mathcal{D}_{t,k}\}$ is i.i.d. Here, $C_{t,k}^{(i)}$ represents the i -th column of $C_{t,k}$, $n_t^{(i)}$ is the total number of samples from class i at stage t , and $n_{t,k}^{(i)}$ is the number of samples from class i for client k at stage t .

Remark 1. The i.i.d. assumption of the global feature set Ω in Proposition 3.1 is valid within the non-i.i.d. setting (data heterogeneity) in FCIL. We explain it in **Appendix A**.

Proposition 3.1 indicates that each client k only needs to transmit the correlation matrix $C_{t,k}$ and the label frequency vector $\mathbf{n}_{t,k}$ ($\mathbf{n}_{t,k} \in \mathbb{R}^{c_t}$) for task t , significantly reducing communication overhead. The server leverages these uploaded statistics to compute the estimated gram matrix $\hat{G}_t$ and performs temporal statistics aggregation (Eq. 3) to derive the accumulated global gram matrix $\hat{G}_{1:t}$. Then $G_{1:t}$ is used in Eq. 7 to update the classifier:

$$\mathbf{W}_t = (\hat{G}_{1:t} + \gamma \mathbf{I})^{-1} \mathbf{C}_{1:t}. \quad (9)$$

Then we provide two key theoretical results to analyze the estimation error of STSA-E in comparison to STSA. The full descriptions and proofs can be found in **Appendix C**.

We consider a noisy linear regression model, where $n = \sum_{i=1}^t n_i$ and $n > M$. Specifically, we assume the existence of a ground-truth weight matrix $\mathbf{W}_t^0 \in \mathbb{R}^{M \times c_{\text{joint}}}$ and noise $\mathcal{E}_{\text{joint}} \in \mathbb{R}^{n \times c_{\text{joint}}}$ such that $\mathbf{Y}_{\text{joint}} = \mathbf{X}_{\text{joint}} \mathbf{W}_t^0 + \mathcal{E}_{\text{joint}}$, where $c_{\text{joint}} = \sum_{i=1}^t c_i$.

For each task $\mathcal{T}_t$, and for every $i \in \{1, \dots, c_t\}$, we make the following assumptions:

Assumption 1 (Data Structure). $\|\mathbf{X}_t\|_F \leq \sqrt{n_t} C_X$, $\text{tr}(\Sigma_t^{(i)}) \leq C_\Sigma$ and $\|\mu_t^{(i)}\|_2^2 \leq C_\mu$ for some constants $C_X, C_\Sigma, C_\mu > 0$, where $\mu_t^{(i)}$ and $\Sigma_t^{(i)}$ are the class-wise mean and covariance of the global feature set Ω at stage t .

Assumption 2 (Class Balance). There exists a constant $0 < C \ll c_t$ such that $c_t \sum_{i=1}^{c_t} n_t^{(i)2} \leq C n_t^2$.

Assumption 3 (Stability). $\|\mathbf{W}_t^0\|_F \leq \sqrt{M} r$ for some constant $r > 0$.

Assumption 4 (Noise). The i -th row of noise satisfies the following conditions: $\mathbb{E}(\mathcal{E}_{\text{joint},i}^\top) = \mathbf{0}$, $\mathbb{E}(\mathcal{E}_{\text{joint},i}^\top \mathcal{E}_{\text{joint},i}) = \sigma^2 \mathbf{I}_M$ and $\mathcal{E}_{\text{joint}}$ is independent of $\mathbf{X}_{\text{joint}}$.

These four assumptions are mild in statistical learning theory and aim to reflect practical conditions without overidealization. We provide a detailed explanation of their rationality in **Appendix C**. **Theorem 3.3 (STSA-E plug-in error).** Under Assumptions 1-2, with high probability, we have

$$\frac{1}{n} \mathbb{E} \|\mathbf{X}_{\text{joint}} \mathbf{W}_t - \mathbf{X}_{\text{joint}} \mathbf{W}^*\|_F \leq O\left(\frac{\frac{K+1}{K-1} t}{n(\frac{\gamma^2}{n^2} + 1)}\right). \quad (10)$$

Remark 2. Theorem 3.3 shows that the plug-in error decreases as the average per-stage sample size $\frac{n}{t}$ increases.

Corollary 3.4 (Estimation error). Under Assumptions 1-4, with high probability, we have

$$\frac{1}{n} \mathbb{E} \|\mathbf{X}_{\text{joint}} \mathbf{W}_t - \mathbf{Y}_{\text{joint}}\|_F \leq O\left(\frac{\frac{K+1}{K-1} t}{n(\frac{\gamma^2}{n^2} + 1)}\right) + O\left(\frac{1}{\sqrt{n}}\right). \quad (11)$$

Remark 3. Corollary 3.4 reveals that our STSA-E achieves $O(n^{-1})$ estimation efficiency, contrasting with the oracle estimator's $O(n^{-1/2})$ rate (Eq. 7) [18]. Thus, STSA-E maintains comparable statistical performance to the STSA with significantly reduced communication overhead, achieving a great performance-resource trade-off.

Dummy Clients. Since the number of clients, K , affects the quality of the estimated $\hat{G}_t$, we introduce dummy clients to improve estimation when K is small. Specifically, we divide each client's local dataset into multiple subsets, treating each subset as a separate dummy client. For instance, client k in task $\mathcal{T}_t$ splits its dataset $\mathcal{D}_{t,k}$ into K_D subsets. Each dummy client k_j ($j = 1, 2, \dots, K_D$) holds a subset $\mathcal{D}_{t,k_j} \subset \mathcal{D}_{t,k}$ and uploads the corresponding spatial statistics C_{t,k_j} and $\mathbf{n}_{t,k_j}$ to the server. This effectively increases the number of clients from K to $K \times K_D$.

Introducing dummy clients brings extra communication overhead for uploading first-order statistics (from $C_{t,k}$ to $\{C_{t,k_j}, \mathbf{n}_{t,k_j}\}_{j=1}^{K_D}$). However, this overhead is typically much smaller than the cost of uploading second-order statistics ($G_{t,k}$) in STSA (Section 4.4). It is worth noting that dummy clients are used to adapt STSA-E to the cross-silo setting, where the number of clients K is small. In the cross-device setting (K is large), they are not needed (Section 4.3).

Figure 3 shows an overview of STSA and Algorithm ?? provides its entire procedure with pseudocode.

4 Experiments

4.1 Experimental Setting

Datasets and Models. We conduct experiments under two settings: training from scratch and training with a pre-trained model. For the scratch setting, we use ResNet18 [16] as the backbone on CIFAR100 [22] and Tiny-ImageNet [23]. For the pre-trained setting, we use ViT-B/16 [11] (pre-trained on ImageNet-21K) as the backbone on CIFAR100 [22] and ImageNet-R [17]. By default, we divide dataset classes into 10 non-overlapping tasks ($T=10$) and distribute them among 5 clients ($K=5$) via a Dirichlet distribution with varying β .

Comparison Baselines. Following the benchmark in [37, 41], we compare our STSA and STSA-E with Fed-EWC [21], Fed-LWF [27], MFCL [3], and LANDER [37] under the training from scratch setting. We also evaluate our method against PEFT-based FCIL methods, including Fed-DualP [39], Fed-CODAP [35], Fed-CPrompt [4], and PILoRA [15], using a pre-trained ViT-B/16. For our method, we integrate an adapter [45] with ViT to ensure performance after the first task is comparable to other PEFT-based methods.

Evaluation Metrics. We employ three widely-used metrics in FCIL [35, 37, 45]: average incremental accuracy (A_{avg}), final average accuracy (A_T), and average forgetting (F_T). A_{avg} denotes the average performance across all incremental stages, A_T is the accuracy averaged over all T tasks after completing task $\mathcal{T}_T$, and F_T represents

Table 1: Experimental results for training from scratch (ResNet18) with varying levels of data heterogeneity. The best and second-best results are shown in bold and underlined, respectively.

Dataset	Data Partition Metrics	$\beta = 1$			$\beta = 0.5$			$\beta = 0.1$		
		$A_{\text{avg}} (\uparrow)$	$A_T (\uparrow)$	$F_T (\downarrow)$	$A_{\text{avg}} (\uparrow)$	$A_T (\uparrow)$	$F_T (\downarrow)$	$A_{\text{avg}} (\uparrow)$	$A_T (\uparrow)$	$F_T (\downarrow)$
CIFAR100	Joint (Upper-bound)	-	76.74	-	-	76.74	-	-	76.74	-
	Fed-EWC [21]	29.23	11.76	73.52	27.16	11.56	70.53	21.21	9.13	60.79
	Fed-LWF [27]	44.89	27.04	44.60	40.25	24.23	42.86	32.28	21.89	38.29
	MFCL [3]	43.48	26.46	31.28	44.00	27.17	27.27	34.48	21.96	22.77
	LANDER [37]	55.01	37.53	28.31	45.68	33.09	35.38	42.41	30.50	26.42
	STSA-E (Ours)	50.29	34.85	9.69	<u>49.16</u>	<u>34.57</u>	9.26	<u>44.86</u>	<u>30.74</u>	<u>9.92</u>
	STSA (Ours)	<u>50.71</u>	<u>35.14</u>	<u>9.81</u>	50.10	34.97	<u>9.57</u>	46.19	31.02	9.73
Tiny-ImageNet	Joint (Upper-bound)	-	49.44	-	-	49.44	-	-	49.44	-
	Fed-EWC [21]	19.49	7.68	52.51	19.07	7.48	52.04	17.55	6.08	49.82
	Fed-LWF [27]	29.11	17.02	27.64	26.91	15.47	28.06	26.54	15.16	25.22
	MFCL [3]	<u>32.58</u>	18.12	24.53	30.35	16.43	28.93	28.21	16.24	36.33
	LANDER [37]	34.53	18.21	38.04	30.75	16.59	26.77	28.49	16.29	33.76
	STSA-E (Ours)	30.86	<u>19.94</u>	<u>8.94</u>	30.91	20.02	<u>7.66</u>	<u>29.07</u>	<u>18.11</u>	<u>8.36</u>
	STSA (Ours)	31.14	20.03	8.26	<u>30.77</u>	<u>19.99</u>	6.56	29.10	18.18	7.50

Table 2: Experimental results for training from pre-trained model (ViT-B/16) with varying levels of data heterogeneity. The best and second-best results are shown in bold and underlined, respectively.

Dataset	Data Partition Metrics	$\beta = 1$			$\beta = 0.5$			$\beta = 0.1$		
		$A_{\text{avg}} (\uparrow)$	$A_T (\uparrow)$	$F_T (\downarrow)$	$A_{\text{avg}} (\uparrow)$	$A_T (\uparrow)$	$F_T (\downarrow)$	$A_{\text{avg}} (\uparrow)$	$A_T (\uparrow)$	$F_T (\downarrow)$
CIFAR100	Joint (Upper-bound)	-	92.45	-	-	92.45	-	-	92.45	-
	Fed-DualP [39]	83.02	75.07	4.78	78.12	65.77	3.97	69.11	56.31	3.31
	Fed-CODAP [35]	82.20	74.12	6.34	74.80	64.02	10.79	67.75	55.54	14.04
	Fed-CPrompt [4]	81.43	73.86	8.29	75.07	65.87	9.28	69.67	61.68	10.03
	PILoRA [15]	83.59	77.17	8.59	83.46	76.81	6.41	82.85	74.09	7.67
	STSA-E (Ours)	<u>93.71</u>	<u>89.42</u>	3.93	<u>93.35</u>	<u>89.37</u>	<u>4.05</u>	<u>93.14</u>	<u>88.65</u>	<u>4.06</u>
	STSA (Ours)	94.05	89.85	<u>4.12</u>	93.96	89.72	4.09	93.63	89.09	4.35
ImageNet-R	Joint (Upper-bound)	-	80.27	-	-	80.27	-	-	80.27	-
	Fed-DualP [39]	56.61	48.58	7.13	53.37	45.25	9.80	35.58	27.92	10.52
	Fed-CODAP [35]	55.10	51.13	6.42	48.49	43.00	6.06	23.22	15.15	8.29
	Fed-CPrompt [4]	51.09	47.28	9.49	45.42	39.12	9.27	23.67	16.87	9.52
	PILoRA [15]	55.47	51.92	6.69	53.43	50.18	7.09	51.85	47.85	7.19
	STSA-E (Ours)	<u>72.89</u>	<u>66.68</u>	<u>6.45</u>	<u>71.77</u>	<u>66.08</u>	<u>6.60</u>	71.67	<u>65.46</u>	6.37
	STSA (Ours)	74.02	67.25	6.84	73.59	66.67	7.14	73.65	66.78	<u>7.18</u>

the average performance decline on previous tasks over all T tasks. Let $A_{t,\tau}$ represent the classification accuracy on the τ -th task after training on the t -th task. After training on the t -th task, we calculate the evaluation metrics as: $A_{\text{avg}} = \sum_{t=1}^T \frac{1}{t} \sum_{\tau=1}^t A_{t,\tau}$, $A_T = \frac{1}{T} \sum_{\tau=1}^T A_{T,\tau}$, $F_T = \frac{1}{T-1} \sum_{\tau=1}^{T-1} \max_{t' \in \{1, \dots, T-1\}} A_{t',\tau} - A_{T,\tau}$.

Experimental Details. For ResNet18, we set the batch size to 128, local training epoch to 2, and communication rounds to 100. For pre-trained ViT-B/16, we use 2 local epochs with a batch size of 128, conducting 10 communication rounds on CIFAR100 and 20 communication rounds on ImageNet-R. The projection dimension of the adapter is set to 64. The dimension of the random feature M is set to 5000 for ResNet18 and 1250 for ViT-B/16. The number of

dummy clients K_D is set to 50 for ResNet18, 10 for ViT-B/16. The ridge coefficient is set to $1e4$ for ResNet18 and $1e6$ for ViT-B/16.

4.2 Main Results

Experiments for training from scratch. We present the results of experiments using ResNet18 in Table 1 and Figure 4. When data heterogeneity is low ($\beta = 1$), the DFKD method (LANDER, MFCL) performs well, and our method is slightly weaker than the best baseline, LANDER. However, as data heterogeneity increases, our method demonstrates greater robustness. On CIFAR100, STSA shows significant performance improvements, particularly in average incremental accuracy. At $\beta = 0.5$, STSA-E outperforms LANDER by 3.48%, and STSA surpasses LANDER by 4.42%. Similarly, on Tiny-ImageNet, STSA shows substantial gains in final incremental

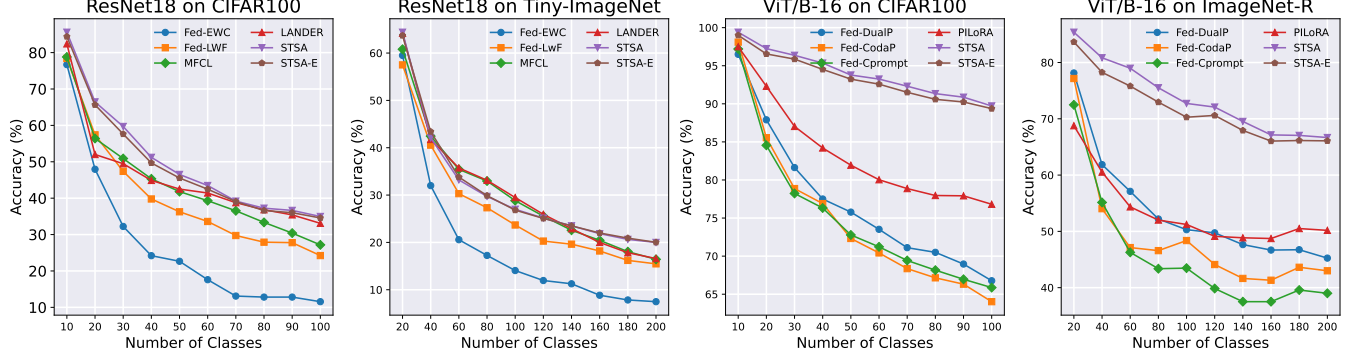


Figure 4: Performance curve of different datasets ($\beta = 0.5$) with different backbone.

accuracy. When $\beta = 0.5$, STSA-E outperforms LANDER by 3.43%, and STSA surpasses LANDER by 3.4%. Under extreme heterogeneity ($\beta = 0.1$), our method, like others, shows some performance drop due to the less effective feature extractor trained in the first stage. This drop can be mitigated by using better federated learning algorithms or a pre-trained model in the first stage.

Experiments for training from pre-trained model. The experimental results of training from pre-trained ViT-B/16 are shown in Table 2. We observe that STSA and STSA-E significantly outperforms the best FCIL baseline, achieving over **10%** improvements in both average incremental accuracy and final average accuracy across all cases. For CIFAR100, our method is just 3% lower than the upper bound. Furthermore, STSA and STSA-E exhibit strong robustness to data heterogeneity, maintaining nearly consistent performance regardless of the degree of heterogeneity. Moreover, we show the performance curve of different methods in Figure 4, which highlights the outstanding performance of our method throughout the entire incremental learning process. Compared to training from scratch, the pre-trained model’s generalizability enables it to learn a strong feature extractor in the first phase, which gives our method a significant performance advantage.

Comparison between STSA and STSA-E. STSA typically slightly outperforms STSA-E, though the reverse can occur, such as on Tiny-ImageNet ($\beta = 0.5$). This is due to estimation errors, which can be seen as random noise that sometimes enhances performance. As noted in several FL studies [42, 43], adding random noise to feature statistics can sometimes improve performance. In summary, STSA-E delivers comparable performance to STSA while significantly reducing communication overhead (see Section 4.4). This makes STSA-E especially suitable for mobile and IoT environments with strict communication limits.

4.3 Ablation Study

Ablation on Dimension of Random Feature. Figure 5 shows the results with different values of M . Here, 512 is the dimension of raw feature in ResNet18, and 768 is the dimension of raw feature in ViT-B/16. We find that random mapping significantly improves the performance of STSA, confirming that random mapping enhances the linear separability of features. Additionally, setting M above 1250 is sufficient to achieve competitive results compared to other state-of-the-art baselines.

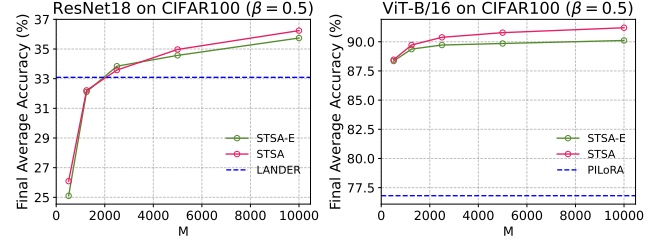


Figure 5: Final average accuracy A_T with different M . The top-performing baselines (LANDER and PiLoRA) are also marked in the figure.

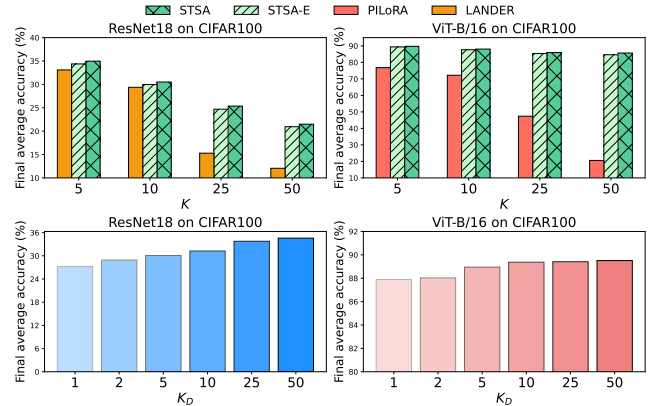


Figure 6: (Top) Final average accuracy with varying client number K . The top-performing baselines (LANDER and PiLoRA) are also marked in the figure. (Bottom) Final average accuracy with varying dummy client number K_D .

Ablation on Number of Clients. The number of clients varied as $K = \{5, 10, 25, 50\}$. For STSA-E, we set K_D to $\{50, 25, 10, 5\}$ for ResNet18, totaling 250 clients ($K \times K_D$), and $\{10, 5, 2, 1\}$ for ViT-B/16, totaling 50 clients. Figure 6 (Top) demonstrates that our method outperforms other top-performing baselines as the number of clients increases. Notably, performance advantage becomes more evident with more clients, as only the first-stage adaptation is influenced by

the number of clients, while the subsequent spatial-temporal statistics aggregation remains unaffected. Meanwhile, when the number of clients is large, there is no need to set dummy clients (ViT with $K = 50, K_D = 1$). Therefore, STSA-E is well-suited for large-scale Web, mobile, and IoT environments where many resource-limited devices collaborate to train models. We also assess the impact of varying the number of dummy clients K_D on the performance of STSA-E. As shown in Figure 6 (Bottom), the performance of STSA-E improves with more dummy clients and gradually approaches that of STSA.

Ablation on First Stage adaptation Strategies. The subsequent statistics aggregation is independent of the first stage adaptation, so our STSA and STSA-E can combine with any PEFT methods for the first stage. Here, we combine two PEFT-based FCIL methods, PiLoRA (LoRA) and Fed-CPrompt (prompt), with ours. Table 3 shows the results. We find that our STSA consistently achieves better performance against other baselines in all metrics. This demonstrates the high flexibility of our method.

PEFT	Metrics	$A_{\text{avg}}(\uparrow)$	$A_T(\uparrow)$	$F_T(\downarrow)$
LoRA	PiLoRA [15]	83.46	76.81	6.41
	STSA-E+PiLoRA	<u>89.45</u>	<u>84.72</u>	5.54
	STSA+PiLoRA	90.51	85.64	<u>5.82</u>
Prompt	Fed-CPrompt [4]	75.07	65.87	9.28
	STSA-E+Fed-CPrompt	90.07	85.67	5.26
	STSA+Fed-CPrompt	89.86	<u>85.60</u>	<u>6.32</u>
Adapter	STSA-E	<u>93.35</u>	<u>89.37</u>	4.05
	STSA	93.96	89.72	<u>4.09</u>

Table 3: Results of different first stage adaptation strategies. The dataset is CIFAR100 ($\beta = 0.5$).

4.4 Further Analysis

Communication Overhead. After first stage adaptation, each client uploads only the Spatial Statistics, represented as $\{G_{t,k}, C_{t,k}\}$ with a size of $(M + c_t) \times M$ for STSA, or $\{C_{t,k_j}, \mathbf{n}_{t,k_j}\}_{j=1}^{K_D}$ with a size of $(M + 1) \times c_t \times K_D$ for STSA-E at stage t . It is especially commendable that only one round of communication happens between the clients and the central server in the subsequent aggregation procedure, which further enhances robustness against potential network disruptions and straggling clients [14]. The left column of Table 4 presents the communication overhead of the entire learning process for each client. For fair comparison, we only evaluate communication overhead after the training in the first stage. Our method significantly reduces communication overhead compared to DFKD methods and achieves results comparable to PEFT methods. Figure 7 shows that our STSA-E significantly reduces communication overhead compared to STSA.

Computation Overhead. Our STSA does not involve the time-consuming backpropagation after the first stage adaptation, which provides a significant advantage in computational efficiency. We measure the total experimental time (including training and testing) after first stage adaptation using a single NVIDIA 3090 RTX GPU. As shown in the right columns in Table 4, our STSA is over 80 times faster than LANDER [37] and 120 times faster than PiLoRA [15].

Method	Communication Overhead ↓	Wallclock Time ↓
ResNet18 on CIFAR100 with 10 tasks, $M = 5000, K_D = 50$		
LANDER [37]	79047.9 MB	> 10 h
STSA-E (Ours)	95.4 MB	< 3 min
STSA (Ours)	955.6 MB	< 3 min
ViT-B/16 on CIFAR100 with 10 tasks, $M = 1250, K_D = 10$		
PiLoRA [15]	36.39 MB	> 20 h
STSA-E (Ours)	4.77 MB	< 10 min
STSA (Ours)	60.1 MB	< 10 min

Table 4: Cost comparison with top-performing baselines after the training in the first stage.

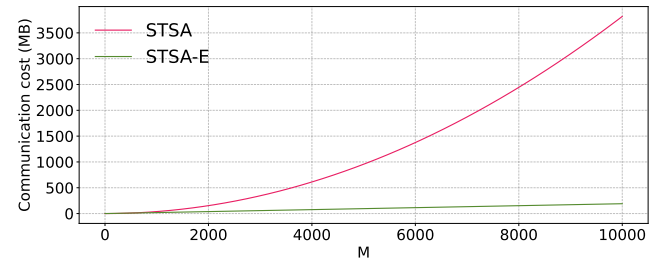


Figure 7: Communication overhead per client after the first stage of training, evaluated with varying M on CIFAR100 ($K_D = 50$).

Privacy of STSA. As same as PiLoRA [15], our method involves uploading feature statistics, which may raise privacy concerns. Specifically, STSA uploads $G_{t,k}$ and $C_{t,k}$, while STSA-E uploads $C_{t,k}$ (assuming no dummy clients are used for simplicity). Feature statistics $G_{t,k}$ and $C_{t,k}$ are the aggregation of feature statistics from all local data samples. In real-world scenarios, each client k typically owns many data samples across various classes, making it difficult to reconstruct specific samples from the uploaded feature statistics. Moreover, compatible privacy-preserving techniques, such as secure aggregation, can be integrated to further strengthen our method's privacy protection, as the server requires only the aggregated Spatial Statistics rather than individual client values.

5 Conclusion

We propose a novel FCIL method, STSA, which unifies feature statistic aggregation across spatial and temporal dimensions. The aggregated statistics are immune to data heterogeneity and enable closed-form classifier updates at each stage. We also introduce STSA-E, a variant that achieves similar performance with lower communication costs. Extensive experiments show that our method outperforms state-of-the-art baselines, offering greater flexibility while reducing communication and computation overhead.

Acknowledgments

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A i.i.d. assumption of global feature set Ω

In FCIL, **non-i.i.d. refers to label skew setting** where the label distributions $P_k(y)$ vary across clients, while the conditional distribution $P_k(x|y)$ remains the same for all clients [37, 41, 47]. Given the shared backbone Φ , the conditional feature distribution $P_k(\Phi(x)|y)$ is also uniform across clients. Therefore, the global feature set $\Omega = \{(\Phi(x), y) \mid (x, y) \in \cup_{k=1}^K \mathcal{D}_k\}$ follows a mixture distribution:

$$P_\Omega(\Phi(X), Y) = \sum_{k=1}^K w_k P_k(\Phi(X), Y) = P(\Phi(X)|Y) \sum_{k=1}^K w_k P_k(Y) \\ = P(\Phi(X)|Y) P_{avg}(Y), \quad (12)$$

where w_k is the proportion of data from client k ($w_k = \frac{|D_k|}{|\cup_{k=1}^K D_k|}$), and $P_{avg}(Y)$ is the global average label distribution across all data.

The non-i.i.d. nature of the clients ($P_k(Y)$) only influences the shape of the final mixture distribution P_Ω (via $P_{avg}(Y)$), but it does not prevent samples drawn from Ω from being identically distributed according to P_Ω . Therefore, the i.i.d. assumption of Ω in Proposition 3.1 (following the single mixture distribution P_Ω) is valid within the label skew setting (non-i.i.d. in FCIL).

This explains why STSA-E shows strong robustness and comparable performance to STSA across varying data heterogeneity in our experiments.

B Proof of Proposition 3.1

Proposition 3.1 For each task $\mathcal{T}_t$,

$$\hat{G}_t = \sum_{i=1}^{c_t} \left[\frac{n_t^{(i)} - 1}{K - 1} \sum_{k=1}^K \frac{C_{t,k}^{(i)} C_{t,k}^{(i)\top}}{n_{t,k}^{(i)}} - \frac{n_t^{(i)} - K}{n_t^{(i)}(K - 1)} \left(\sum_{k=1}^K C_{t,k}^{(i)} \right) \left(\sum_{k=1}^K C_{t,k}^{(i)} \right)^\top \right], \quad (13)$$

is an unbiased plug-in estimator of the global gram matrix G_t when global feature set $\Omega = \{(\Phi(x), y) \mid (x, y) \in \cup_{k=1}^K \mathcal{D}_{t,k}\}$ is i.i.d. Here, $C_{t,k}^{(i)}$ denotes the i -th column of $C_{t,k}$ and $n_t^{(i)}$ denotes the total number of data samples from class i in stage t , and $n_{t,k}^{(i)}$ denotes the number of samples from class i for client k at stage t .

PROOF. For simplicity, we omit the subscript t in the proof. Under the i.i.d assumption of the global feature set $\{(\Phi(x), y) \mid (x, y) \in \cup_{k=1}^K D_k\}$, and letting $n_k^{(i)}$ denote the total number of data samples from class i in client k , we have $\mathbb{E}[C_k^{(i)}] = \sum_{(x,i) \in D_k} \mathbb{E}[\Phi(x)] := n_k^{(i)} \mu^{(i)}$ and $\mathbb{D}[C_k^{(i)}] = \sum_{(x,i) \in D_k} \mathbb{D}[\Phi(x)] := n_k^{(i)} \Sigma^{(i)}$.

Here, $\mu^{(i)}$ and $\Sigma^{(i)}$ are the class-wise mean and covariance of the global feature set Ω . Since $G = \sum_{i=1}^c G^{(i)} = \sum_{i=1}^c \sum_{k=1}^K G_k^{(i)}$,

we have $\mathbb{E}[G] = \sum_{i=1}^c \mathbb{E}[G^{(i)}]$ and

$$\mathbb{E}[G^{(i)}] = \sum_{k=1}^K \mathbb{E}[G_k^{(i)}] = \sum_{k=1}^K \mathbb{E}[X_k^{(i)\top} X_k^{(i)}] \\ = \sum_{k=1}^K (n_k^{(i)} \mu^{(i)} \mu^{(i)\top} + n_k^{(i)} \Sigma^{(i)}) = n^{(i)} \mu^{(i)} \mu^{(i)\top} + n^{(i)} \Sigma^{(i)} \quad (14)$$

For clarity in the proof, we expand $\mathbb{E}[\hat{G}]$ and divide it into several parts as follows:

$$\mathbb{E}[\hat{G}] = \sum_{i=1}^c \mathbb{E}[\hat{G}^{(i)}] = \sum_{i=1}^c \left[\frac{n^{(i)} - 1}{K - 1} A_{1,i} - \frac{n^{(i)} - K}{n^{(i)}(K - 1)} A_{2,i} \right],$$

$$\text{where } A_{1,i} = \sum_{k=1}^K \frac{\mathbb{E}[C_k^{(i)} C_k^{(i)\top}]}{n_k^{(i)}}, A_{2,i} = \mathbb{E} \left[\left(\sum_{k=1}^K C_k^{(i)} \right) \left(\sum_{k=1}^K C_k^{(i)} \right)^\top \right]. \quad (15)$$

To proceed, we compute each term in above equation separately:

$$A_{1,i} = \sum_{k=1}^K \frac{\mathbb{E}[C_k^{(i)}] \mathbb{E}[C_k^{(i)}]^\top + \mathbb{D}[C_k^{(i)}]}{n_k^{(i)}} \\ = \sum_{k=1}^K (n_k^{(i)} \mu^{(i)} \mu^{(i)\top} + \Sigma^{(i)}) = n^{(i)} \mu^{(i)} \mu^{(i)\top} + K \Sigma^{(i)} \quad (16)$$

$$A_{2,i} = \mathbb{E} \left[\sum_{k=1}^K C_k^{(i)} \right] \mathbb{E} \left[\sum_{k=1}^K C_k^{(i)} \right]^\top + \mathbb{D} \left[\sum_{k=1}^K C_k^{(i)} \right] \\ = \left(\sum_{k=1}^K n_k^{(i)} \mu^{(i)} \right) \left(\sum_{k=1}^K n_k^{(i)} \mu^{(i)} \right)^\top + \sum_{k=1}^K n_k^{(i)} \Sigma^{(i)} \\ = (n^{(i)})^2 \mu^{(i)} \mu^{(i)\top} + n^{(i)} \Sigma^{(i)} \quad (17)$$

Therefore, by substituting $A_{1,i}$ and $A_{2,i}$ back into the equation for $\mathbb{E}[\hat{G}]$, we get:

$$\mathbb{E}[\hat{G}] = \sum_{i=1}^c \left[\frac{n^{(i)} - 1}{K - 1} (n^{(i)} \mu^{(i)} \mu^{(i)\top} + K \Sigma^{(i)}) - \frac{n^{(i)} - K}{n^{(i)}(K - 1)} ((n^{(i)})^2 \mu^{(i)} \mu^{(i)\top} + n^{(i)} \Sigma^{(i)}) \right] \\ = \sum_{i=1}^c \left[n^{(i)} \mu^{(i)} \mu^{(i)\top} + n^{(i)} \Sigma^{(i)} \right] = \sum_{i=1}^c \mathbb{E}[G^{(i)}] = \mathbb{E}[G] \quad (18)$$

□

C Estimation Guarantee of STSA-E

Notation: Let $\|\cdot\|_2$ denote the spectral norm, $\|\cdot\|_F$ represent the Frobenius norm, $\mathbb{E}$ indicate the expectation operator, $\mathbb{D}$ signify the variance operator, $\mathbf{0}$ be the zero vector, $\mathbf{I}_M$ denote the M -dimensional identity matrix, and $\text{tr}(\cdot)$ represent the matrix trace operation.

We consider a noisy linear regression model and let n represent the total number of collected samples until stage t , that is, $n = \sum_{i=1}^t n_i$ and $c_{\text{joint}} = \sum_{i=1}^t c_i$. In the subsequent discussion, we assume that, through sufficient continual learning and accumulation, the number of samples n exceeds the feature dimension M ,

i.e., $n > M$. Specifically, we assume there exist ground-truth weight matrix $\mathbf{W}_t^0 \in \mathbb{R}^{M \times c_{\text{joint}}}$ and noise $\mathcal{E}_{\text{joint}} \in \mathbb{R}^{n \times c_{\text{joint}}}$ satisfying

$$\mathbf{Y}_{\text{joint}} = \mathbf{X}_{\text{joint}} \mathbf{W}_t^0 + \mathcal{E}_{\text{joint}}. \quad (19)$$

For each task $\mathcal{T}_t$, and for every $i \in \{1, \dots, c_t\}$, we assume:

Assumption 1 (Data Structure). $\|\mathbf{X}_t\|_F \leq \sqrt{n_t} C_X$, $\text{tr}(\Sigma_t^{(i)}) \leq C_\Sigma$ and $\|\mu_t^{(i)}\|_2^2 \leq C_\mu$ for some constants $C_X, C_\Sigma, C_\mu > 0$, where $\mu_t^{(i)}$ and $\Sigma_t^{(i)}$ are the class-wise mean and covariance of the global feature set Ω at stage t .

Assumption 2 (Class Balance). There exists a constant $0 < C \ll c_t$ such that $c_t \sum_{i=1}^{c_t} n_t^{(i)2} \leq C n_t^2$.

Assumption 3 (Stability). $\|\mathbf{W}_t^0\|_F \leq \sqrt{M}r$ for some constant $r > 0$.

Assumption 4 (Noise). The i -th row of noise satisfies the following conditions: $\mathbb{E}(\mathcal{E}_{\text{joint},i}^\top) = \mathbf{0}$, $\mathbb{E}(\mathcal{E}_{\text{joint},i}^\top \mathcal{E}_{\text{joint},i}) = \sigma^2 \mathbf{I}_M$ and $\mathcal{E}_{\text{joint}}$ is independent of $\mathbf{X}_{\text{joint}}$.

We formalize these assumptions to ensure theoretical rigor while maintaining practical relevance. Here, we further explain the rationality of these assumptions:

Assumptions 1 (Data Structure) and 3 (Stability) are common boundedness conditions in theoretical analyses [33]. They simply ensure that feature and weight norms remain finite, which naturally holds in real-world scenarios. **Assumption 2 (Class Balance)** is a mild and flexible condition for global class distribution (considering all data across all clients): it does not require equal sample sizes across classes but only rules out extreme imbalance where one class dominates almost all samples (i.e., $\sum n_t^{(i)2} \ll n_t^2$). This aligns with practical FCIL settings, where the global class distribution typically includes multiple classes with reasonable diversity. **Assumption 4 (Noise)** generalizes standard noise models in statistical learning. We only require zero-mean, homoscedastic, and feature-independent noise. This condition is less restrictive than many prior works that assume Gaussianity [8, 33], enabling broader applicability to real-world noise patterns without imposing stringent distributional constraints.

Proposition 3.2. Under Assumption 1-2, when considering $\hat{\mathbf{G}}_t$ as described in Proposition 3.1, we obtain the following results.

$$\mathbb{E} \|\hat{\mathbf{G}}_t - \mathbf{G}_t\|_F^2 \leq C' C_\Sigma^2 \left(\frac{K+1}{K-1}\right)^2 n_t^2 = O\left(\left(\frac{K+1}{K-1}\right)^2 n_t^2\right), \quad (20)$$

where C' represents a constant that is independent of (n_t, K) .

PROOF. For simplicity, we omit the subscript t in the proof.

$$\begin{aligned} \|\hat{\mathbf{G}} - \mathbf{G}\|_F^2 &\leq c \sum_{i=1}^c \|\hat{\mathbf{G}}^{(i)} - \mathbf{G}^{(i)}\|_F^2 \\ &= c \sum_{i=1}^c \left\| \frac{n^{(i)} - 1}{K-1} \sum_{k=1}^K \frac{\mathbf{C}_k^{(i)} \mathbf{C}_k^{(i)\top}}{n_k^{(i)}} - \frac{n^{(i)} - K}{n^{(i)}(K-1)} \left(\sum_{k=1}^K \mathbf{C}_k^{(i)} \right) \left(\sum_{k=1}^K \mathbf{C}_k^{(i)} \right)^\top \right. \\ &\quad \left. - \sum_{k=1}^K \mathbf{X}_k^{(i)\top} \mathbf{X}_k^{(i)} \right\|_F^2 \\ &\leq (K+1)c \sum_{i=1}^c \left(\left(\frac{n^{(i)} - K}{K-1}\right)^2 A_1^{(i)} + \left(\frac{n^{(i)} - 1}{K-1}\right)^2 \sum_{k=1}^K A_{2,k}^{(i)} \right), \end{aligned}$$

where $A_1^{(i)}$ measures global random error, $A_{2,k}^{(i)}$ measures local random error,

$$\begin{aligned} A_1^{(i)} &= \left\| \mathbf{X}^{(i)\top} \mathbf{X}^{(i)} - \frac{1}{n^{(i)}} \left(\sum_{k=1}^K \mathbf{C}_k^{(i)} \right) \left(\sum_{k=1}^K \mathbf{C}_k^{(i)} \right)^\top \right\|_F^2, \\ A_{2,k}^{(i)} &= \left\| \mathbf{X}_k^{(i)\top} \mathbf{X}_k^{(i)} - \frac{1}{n_k^{(i)}} \mathbf{C}_k^{(i)} \mathbf{C}_k^{(i)\top} \right\|_F^2. \end{aligned} \quad (21)$$

According to Lemma 3.5, we have

$$\mathbb{E}[A_1^{(i)}] \leq \frac{n^{(i)} - 1}{n^{(i)}} \text{tr}(\Sigma^{(i)})^2, \quad \mathbb{E}[A_{2,k}^{(i)}] \leq \frac{n_k^{(i)} - 1}{n_k^{(i)}} \text{tr}(\Sigma^{(i)})^2, \quad (22)$$

Substituting them into Eq. C, we have:

$$\begin{aligned} \mathbb{E} \|\hat{\mathbf{G}} - \mathbf{G}\|_F^2 &\leq C_\Sigma^2 (K+1)c \sum_{i=1}^c \left(\left(\frac{n^{(i)} - K}{K-1}\right)^2 \frac{n^{(i)} - 1}{n^{(i)}} + \left(\frac{n^{(i)} - 1}{K-1}\right)^2 \sum_{k=1}^K \frac{n_k^{(i)} - 1}{n_k^{(i)}} \right) \\ &\leq C_\Sigma^2 \frac{(K+1)^2}{(K-1)^2} c \sum_{i=1}^c n^{(i)2} \leq C' C_\Sigma^2 \frac{(K+1)^2}{(K-1)^2} n^2 = O\left(\left(\frac{K+1}{K-1}\right)^2 n^2\right), \end{aligned} \quad (23)$$

where we utilize our class balanced condition to get another constant C' . $\square$

Remark. Proposition 3.2 indicates that the error between $\hat{\mathbf{G}}_t$ and $\mathbf{G}_t$ is highly sensitive to small values of K , but the sensitivity diminishes as K increases.

After quantitatively analyzing the relationship between $\hat{\mathbf{G}}_t$ and $\mathbf{G}_t$, we can use Proposition 3.1 and 3.2 to further explore more profound theoretical properties between $\mathbf{W}_t = (\hat{\mathbf{G}}_{1:t} + \gamma \mathbf{I})^{-1} \mathbf{C}_{1:t}$ and $\mathbf{W}^* = (\mathbf{G}_{1:t} + \gamma \mathbf{I})^{-1} \mathbf{C}_{1:t}$.

Theorem 3.3 (STSA-E plug-in error). Under Assumption 1-2, with high probability, we have

$$\begin{aligned} \frac{1}{n} \mathbb{E} \|\mathbf{X}_{\text{joint}} \mathbf{W}_t - \mathbf{X}_{\text{joint}} \mathbf{W}^*\|_F &\leq \frac{C' C_X (C_\Sigma)^{3/2} C_\mu^{1/2} \frac{K+1}{K-1} t}{n \left(\frac{\gamma^2}{n^2} + C' \right)} = O\left(\frac{\frac{K+1}{K-1} t}{n \left(\frac{\gamma^2}{n^2} + 1 \right)} \right), \end{aligned} \quad (24)$$

where C', C'' are the constants that are independent of (n, t, K, γ) .

PROOF. Firstly, we expand the term in expectation:

$$\begin{aligned} \|\mathbf{X}_{\text{joint}} \mathbf{W}_t - \mathbf{X}_{\text{joint}} \mathbf{W}^*\|_F &\leq \|\mathbf{X}_{\text{joint}}\|_F \|\mathbf{W}_t - \mathbf{W}^*\|_F \\ &= \|\mathbf{X}_{\text{joint}}\|_F \left\| \left(\hat{\mathbf{G}}_{1:t} + \gamma \mathbf{I} \right)^{-1} - \left(\mathbf{G}_{1:t} + \gamma \mathbf{I} \right)^{-1} \right\|_F \|\mathbf{C}_{1:t}\|_F \\ &= \|\mathbf{X}_{\text{joint}}\|_F \left\| \left(\hat{\mathbf{G}}_{1:t} + \gamma \mathbf{I} \right)^{-1} (\mathbf{G}_{1:t} - \hat{\mathbf{G}}_{1:t}) (\mathbf{G}_{1:t} + \gamma \mathbf{I})^{-1} \mathbf{C}_{1:t} \right\|_F \\ &\leq \|\mathbf{X}_{\text{joint}}\|_F \left\| \left(\hat{\mathbf{G}}_{1:t} + \gamma \mathbf{I} \right)^{-1} \right\|_2 \|\mathbf{G}_{1:t} - \hat{\mathbf{G}}_{1:t}\|_F \left\| (\mathbf{G}_{1:t} + \gamma \mathbf{I})^{-1} \right\|_2 \|\mathbf{C}_{1:t}\|_F \\ &\leq \frac{\|\mathbf{X}_{\text{joint}}\|_F \|\mathbf{C}_{1:t}\|_F \|\mathbf{G}_{1:t} - \hat{\mathbf{G}}_{1:t}\|_F}{\gamma^2 + C' n^2} \leq \frac{C_X \sqrt{tn} \|\mathbf{C}_{1:t}\|_F}{\gamma^2 + C' n^2} \sum_{i=1}^t \|\mathbf{G}_i - \hat{\mathbf{G}}_i\|_F, \end{aligned} \quad (25)$$

Regarding the third inequality, we consider $\|(G_{1:t} + \gamma I)^{-1}\|_2$ firstly. Notice that

$$\|(G_{1:t} + \gamma I)^{-1}\|_2 = \frac{1}{\min_i \lambda_i + \gamma}. \quad (26)$$

Where λ_i is the i -th eigenvalue of $G_{1:t}$ and $G_{1:t}$ is the sum of the outer products of n vectors. According to [38] (Theorem 6.1), when the number of vectors n exceeds the dimension M , with high probability, we can derive that $\min_i \lambda_i \geq O(n)$. Therefore, we can postulate the existence of a certain positive constant C' such that:

$$\|(G_{1:t} + \gamma I)^{-1}\|_2 \cdot \left\| \left(\hat{G}_{1:t} + \gamma I \right)^{-1} \right\|_2 \leq \frac{1}{C'n^2 + \gamma^2}. \quad (27)$$

Regarding the fourth inequality, it is derived by considering the Cauchy inequality and the fact that $\|X_{\text{joint}}\|_F \leq \sum_{i=1}^t \|X_i\|_F$, where X_{joint} is the matrix obtained by concatenating the matrices X_i .

By Cauchy inequality, we have

$$\begin{aligned} \mathbb{E}[\|C_{1:t}\|_F \sum_{i=1}^t \|G_i - \hat{G}_i\|_F] &= \sum_{i=1}^t \mathbb{E}[\|C_{1:t}\|_F \|G_i - \hat{G}_i\|_F] \\ &\leq \sum_{i=1}^t \sqrt{\mathbb{E}[\|C_{1:t}\|_F^2] \mathbb{E}[\|G_i - \hat{G}_i\|_F^2]} \end{aligned} \quad (28)$$

Then, we bound $\mathbb{E}[\|C_{1:t}\|_F^2]$,

$$\begin{aligned} \mathbb{E}[\|C_{1:t}\|_F^2] &\leq \mathbb{E}[\sum_{i=1}^t \|C_i\|_F^2] \leq \mathbb{E}[\sum_{i=1}^t \|C_i\|_F^2] \leq t \sum_{i=1}^t \mathbb{E}[\|C_i\|_F^2] \\ &= t \sum_{i=1}^t \mathbb{E}[\text{tr}(X_i^\top Y_i Y_i^\top X_i)] = t \sum_{i=1}^t \mathbb{E}[\text{tr}(Y_i Y_i^\top X_i X_i^\top)] \\ &= t \sum_{i=1}^t \sum_{j=1}^{c_i} n_i^{(j)} \left(\mu_i^{(j)\top} \mu_i^{(j)} + \text{tr}(\Sigma^{(j)}) \right) \leq C_\Sigma C_\mu t n. \end{aligned} \quad (29)$$

By taking the result in Proposition 3.2, and combining the above parts, we have

$$\begin{aligned} \mathbb{E}[\|X_{\text{joint}} W_t - X_{\text{joint}} W^*\|_F] &\leq \frac{C_X C_\Sigma^{1/2} C_\mu^{1/2} t}{\left(\frac{\gamma^2}{n} + C'n\right)} \sum_{i=1}^t \sqrt{\mathbb{E}[\|G_i - \hat{G}_i\|_F^2]} \\ &\leq \frac{C'' C_X (C_\Sigma)^{3/2} C_\mu^{1/2} \frac{K+1}{K-1} t}{\frac{\gamma^2}{n^2} + C'}, \end{aligned} \quad (30)$$

where C'' is the constant in Proposition 3.2. $\square$

Remark. Theorem 3.3 establishes that the plug-in error decreases as the average per-stage sample size $\frac{n}{t}$ increases.

Corollary 3.4 (Estimation error). Under the Assumption 1-4, with high probability, we have

$$\frac{1}{n} \mathbb{E}[\|X_{\text{joint}} W_t - Y_{\text{joint}}\|_F] \leq \frac{C'' C_X (C_\Sigma)^{3/2} C_\mu^{1/2} \frac{K+1}{K-1} t}{n \left(\frac{\gamma^2}{n^2} + C'\right)} + \frac{MRC_X + \sigma}{\sqrt{n}}, \quad (31)$$

where C', C'' are the constants that are independent of (n, t, K, γ) .

PROOF. Consider the error decomposition:

$$\begin{aligned} \frac{1}{n} \mathbb{E}[\|X_{\text{joint}} W_t - Y_{\text{joint}}\|_F] &\leq \frac{1}{n} \mathbb{E}[\|X_{\text{joint}} W_t - X_{\text{joint}} W^*\|_F] + \frac{1}{n} \mathbb{E}[\|X_{\text{joint}} W^* - Y_{\text{joint}}\|_F] \\ &\leq \underbrace{\frac{1}{n} \mathbb{E}[\|X_{\text{joint}} W_t - X_{\text{joint}} W^*\|_F]}_{\text{part1}} + \underbrace{\frac{1}{n} \sqrt{\mathbb{E}[\|X_{\text{joint}} W^* - Y_{\text{joint}}\|_F^2]}}_{\text{part2}}, \end{aligned} \quad (32)$$

where the first term has been analyzed in Theorem 3.3, and we introduce Lemma 3.6 to analyze the second term. Then

$$\text{part1} \leq \frac{C'' C_X (C_\Sigma)^{3/2} C_\mu^{1/2} \frac{K+1}{K-1} t}{n \left(\frac{\gamma^2}{n^2} + C'\right)} = O\left(\frac{\frac{K+1}{K-1} t}{n \left(\frac{\gamma^2}{n^2} + 1\right)}\right). \quad (33)$$

$$\text{part2} \leq \frac{MRC_X + \sigma}{\sqrt{n}} = O\left(\frac{1}{\sqrt{n}}\right). \quad (34)$$

Combing this two parts, we can proof the Corollary 3.4. $\square$

Remark. The most significant implication of Corollary 3.4 lies in the rate analysis: our STSA-E incurs an estimation efficiency loss of order $O(tn^{-1})$, whereas the oracle estimator achieves the standard rate $O(n^{-1/2})$. Crucially, when $t \ll \sqrt{n}$ (a mild scaling condition), the efficiency loss becomes asymptotically negligible. Thus, our method achieves comparable statistical performance to the oracle approach while requiring substantially lower communication overhead, establishing a favorable trade-off between precision and resource efficiency.

Lemma 3.5 ([5]). Let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ be i.i.d. random vectors such that $\mathbf{x}_i \sim \mathcal{N}(\mu, \Sigma)$ for $i = 1, 2, \dots, n$, where $\Sigma = (\sigma_{ij})_{i,j=1}^d$ is a $d \times d$ positive covariance matrix. Define the matrix $\mathbf{M} = \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top - \frac{1}{n} (\sum_{i=1}^n \mathbf{x}_i) (\sum_{i=1}^n \mathbf{x}_i)^\top$. Then, the expected value of the Frobenius norm of $\mathbf{M}$ satisfies the following equality:

$$\mathbb{E}[\|\mathbf{M}\|_F^2] = \frac{n-1}{n} \text{tr}(\Sigma)^2.$$

Lemma 3.6 ([18]). Suppose that $Y = XW_0 + \epsilon$, where $Y \in \mathbb{R}^{n \times c}$, $X \in \mathbb{R}^{n \times M}$ is a random matrix whose rows $\mathbf{x}_i \sim \mathcal{N}(\mu, \Sigma)$ for $i = 1, \dots, n$, $W_0 \in \mathbb{R}^{M \times c}$, and $\epsilon \in \mathbb{R}^{n \times c}$ is a noise matrix with $\mathbb{E}(\epsilon) = \mathbf{0}$, $\mathbb{E}(\epsilon \epsilon^\top) = \sigma^2 \mathbf{I}_n$, and ϵ is independent of X . Let $W = (X^\top X + \gamma I_M)^{-1} X^\top Y$ be the ridge estimator. Then

$$\mathbb{E}[\|XW - Y\|_F^2] = \underbrace{\mathbb{E}[\text{tr}[(H - I_n)^2 XW_0 W_0^\top X^\top]}_{\text{Bias}} + \underbrace{\text{tr}[\mathbb{E}(H - I_n)^2]}_{\text{Variance}} \sigma^2. \quad (35)$$

where $H = X(X^\top X + \gamma I_M)^{-1} X^\top$. Besides, suppose further that: (1). The covariate matrix X satisfies $\|X\|_F \leq \sqrt{n} C_X$ for some constant $C_X > 0$. (2). The true parameter matrix W_0 satisfies $\|W_0\|_F \leq \sqrt{MR}$ for some constant $R > 0$. Then, we have:

$$\mathbb{E}[\|XW - Y\|_F^2] \leq n(M^2 R^2 C_X^2 + \sigma^2). \quad (36)$$